

PackLab: reproducible Percus–Yevick correlations for hard-sphere mixtures

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Abstract

PackLab is an open-source Python package centred on the analytical Percus–Yevick (PY) solution for multicomponent hard-sphere mixtures. It evaluates equilibrium-reference partial pair correlations and structure factors with unit-aware inputs and explicit Fourier-resolution diagnostics. A qualitative comparison with published real-space curves recovers the expected contact locations, curve ordering, and post-contact structure, while an external reciprocal-space implementation confirms mono-, bi-, and ternary-mixture results to within 9.6×10^{-10} . Compiled random sequential addition (RSA) tools complement the analytical workflow by constructing non-overlapping configurations, while a complementary Metropolis Monte Carlo (MMC) utility evolves valid configurations through fixed-volume trial moves.

Keywords: hard-sphere mixtures, Percus–Yevick, pair correlation, structure factor, random sequential addition, Monte Carlo

1. Motivation and significance

Hard-sphere-mixture correlation functions and structure factors are needed as reproducible structural inputs for colloids, particulate materials, porous media, disordered optical systems, and related scattering models [1, 2]. Reproducible studies require varying radius distributions, compositions, and

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Table 1: Code metadata.

Nr.	Code metadata description	Metadata
C1	Current code version	0.7.0
C2	Permanent link to code/repository used for this version	https://github.com/MartinPdeS/PackLab/tree/v0.7.0
C3	Permanent link to reproducible capsule	Not available; validation inputs and figure-generation scripts are versioned with the source.
C4	Legal code license	MIT License
C5	Code versioning system used	Git
C6	Software code languages, tools, and services used	Python, C++20, CMake, pybind11, NumPy, Matplotlib, and TypedUnit
C7	Compilation requirements, operating environments, and dependencies	Python 3.10 or later; CMake; a C++20 compiler; pybind11; OpenMP; NumPy, Matplotlib, and TypedUnit
C8	Link to developer documentation/manual	https://martinpdes.github.io/PackLab/docs/latest/index.html
C9	Support email for questions	martin.poinsinet.de.sivry@gmail.com

Table 2: Executable software metadata.

Nr.	Software metadata description	Metadata
S1	Current software version	0.7.0
S2	Permanent link to executables of this version	https://pypi.org/project/PackLab/0.7.0/ ; https://anaconda.org/martinpdes/packlab
S3	Permanent link to reproducible capsule	Not available; the tagged source and versioned validation data reproduce the reported calculations.
S4	Legal software license	MIT License
S5	Computing platforms/operating systems	Linux, macOS, and Microsoft Windows
S6	Installation requirements and dependencies	<code>pip install PackLab</code> ; Python 3.10 or later; prebuilt wheels target CPython 3.11–3.13
S7	Link to user manual	https://martinpdes.github.io/PackLab/docs/latest/index.html
S8	Support email for questions	martin.poinsinet.de.sivry@gmail.com

⁶ volume fractions while maintaining consistent physical units and Fourier-grid
⁷ resolution.

⁸ Dependent-scattering calculations for dense, polydisperse suspensions provide

9 a concrete example of this need. Attendu et al. combine single-particle Mie
 10 theory with a multicomponent Percus–Yevick (PY) hard-sphere model to
 11 predict the scattering properties of Intralipid across wavelength and concen-
 12 tration [3]. Such applications require the particle-size distribution, mixture
 13 correlation model, and numerical settings to remain internally consistent.
 14 PackLab provides this analytical reference in one documented Python interface.
 15 Its core `analytical` namespace evaluates the PY hard-sphere-mixture model
 16 [4, 5], retaining the mixture definition, dimensional inputs, and Fourier-
 17 resolution choices with the result. The package is therefore suited to analytical
 18 parameter sweeps and structure-aware scattering models. Complementary
 19 random sequential addition (RSA) and Metropolis Monte Carlo (MMC)
 20 workflows provide explicit configurations for downstream coordinate-based
 21 calculations and carefully interpreted comparisons; they are not substitutes
 22 for the equilibrium analytical reference.

23 *1.1. State of the field*

24 Existing software overlaps with individual parts of PackLab rather than with
 25 its complete workflow. SasView is a broader small-angle-scattering analysis
 26 environment that includes monodisperse and binary hard-sphere models using
 27 the PY closure [6, 7]. The mode-coupling package `mctpy` also contains an
 28 `hsmPY` component that evaluates reciprocal-space PY structure factors for
 29 mixtures specified by number densities and diameters [8]. This component
 30 overlaps with one PackLab output, but `mctpy`’s broader purpose and interface
 31 are different.

32 PackLab also supports scattering workflows through its optional PyMieSim
 33 integration, but its specific contribution is to keep multicomponent mixture
 34 construction, real-space partial pair correlations, reciprocal-space partial
 35 structure factors, physical units, and Fourier-grid diagnostics within one
 36 Python workflow. Its explicit-configuration tools additionally make it possi-
 37 ble to generate RSA packings, evolve valid configurations with hard-sphere
 38 MMC, and compare their estimated correlations with the analytical reference
 39 while preserving the distinction between the underlying models. PackLab is
 40 therefore intended for reproducible programmatic studies in which mixture
 41 definitions and numerical resolution must remain traceable across analytical,
 42 configuration, and optional scattering calculations; it is not a replacement for
 43 a general scattering-fitting or mode-coupling environment.

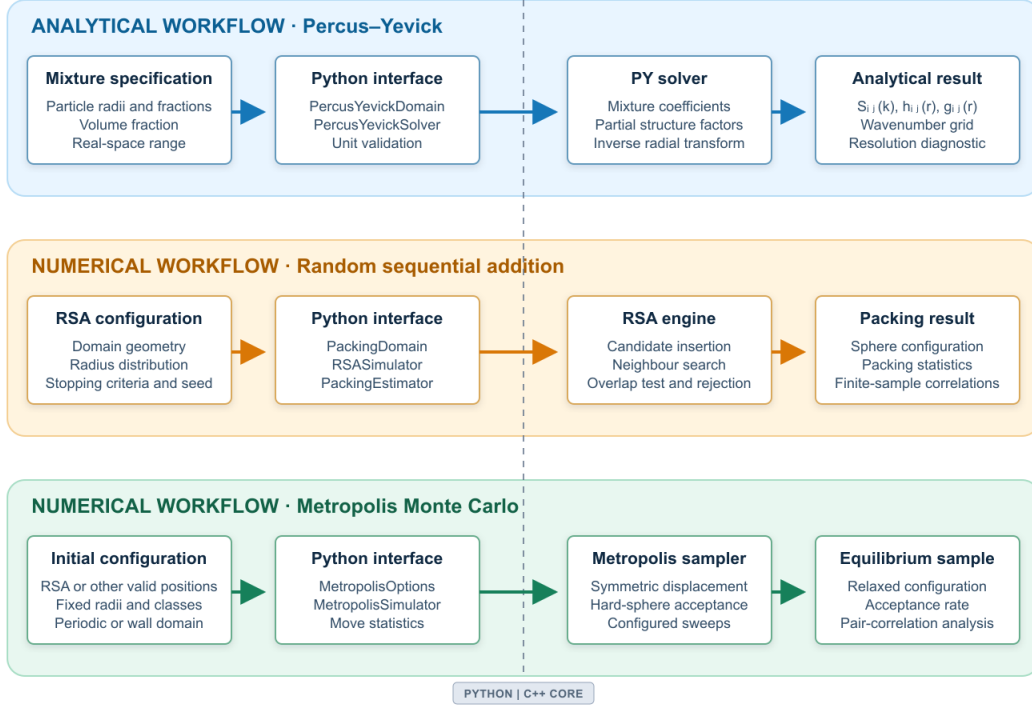


Figure 1: PackLab centres the PY equilibrium hard-sphere-mixture workflow and provides complementary numerical workflows. The analytical workflow evaluates pair correlations and structure factors for specified mixture inputs. The RSA and MMC workflows create irreversible deposition configurations and evolve valid configurations through fixed-volume trial moves, respectively. All workflows have unit-aware Python APIs backed by native C++ kernels, but their results are compared and interpreted rather than treated as interchangeable.

2. Software description

Figure 1 summarises the project structure and the three complementary workflows: the analytical PY model, the irreversible RSA generator, and the MMC sampler.

2.1. Percus–Yevick hard-sphere mixtures

The analytical workflow evaluates the multicomponent PY approximation for hard-sphere mixtures. `PercusYevickDomain` converts radii, number fractions, and total volume fraction into species densities. `PercusYevickSolver` then

52 evaluates partial pair correlations $g_{ij}(r)$ and structure factors on a user-
 53 supplied or automatically constructed wavenumber grid.

54 PY is an approximate equilibrium reference. Its approximation error depends
 55 on mixture density, composition, and size ratio, while its numerical error de-
 56 pends separately on the inverse-transform grid. PackLab exposes both choices:
 57 automatic grids provide a documented default and the grid-construction in-
 58 terface permits explicit resolution studies. The digitised binary-mixture PY
 59 curves of Tsang et al. [9] provide an external visual comparison at matched
 60 radii and partial volume fractions.

61 *2.2. Complementary simulation workflows*

62 *2.2.1. Random sequential addition*

63 RSA proposes one sphere at a time. For a candidate centre $\mathbf{x}_i$ and radius a_i ,
 64 PackLab accepts the candidate only if

$$\|\mathbf{x}_i - \mathbf{x}_j\| \geq a_i + a_j \quad (1)$$

65 for every previously accepted sphere j . Periodic simulations use the minimum-
 66 image convention. For a domain with volume V , the realised packing fraction
 67 is

$$\phi = \frac{1}{V} \sum_i \frac{4\pi a_i^3}{3}. \quad (2)$$

68 RSA is irreversible and history dependent: once a sphere is accepted, it
 69 remains fixed for the rest of the run, while rejected candidates are discarded
 70 and never reconsidered. Consequently, early accepted spheres can block later
 71 positions permanently, so the final configuration depends on the order of
 72 random proposals rather than being a sample from the equilibrium hard-
 73 sphere fluid [10]. PackLab makes this protocol explicit through independent
 74 stopping conditions, a stored random seed, periodic or finite domains, and
 75 constant, uniform, normal, log-normal, and discrete radius samplers. A spatial-
 76 neighbour grid limits overlap checks to nearby accepted spheres, preserving
 77 the RSA acceptance rule while avoiding an all-pairs test for every proposal.

78 *2.2.2. Metropolis Monte Carlo*

79 The MMC utility takes a valid non-overlapping configuration, holds its particle
 80 count, radii, and class labels fixed, and proposes symmetric single-particle
 81 displacements. A proposal is accepted only when it remains within the

domain and introduces no overlap, and the result reports displacement-move acceptance statistics. Users remain responsible for assessing burn-in, autocorrelation, and finite-size effects before interpreting saved states as equilibrium samples. The resulting trajectory can be checked against the matched PY reference through independently initialized chains and an explicit burn-in diagnostic.

2.3. Software design and functionality

Performance-critical analytical and configuration kernels are implemented in C++20 and exposed as native Python extensions. The Python layer provides NumPy-compatible results, `TypedUnit`-based dimensional inputs, plotting helpers, type information, and executable examples. Native extensions are installed beside their public Python APIs, keeping imports such as `PackLab.monte_carlo` and `PackLab.analytical` stable and discoverable. An analytical workflow first defines a mixture with `PercusYevickDomain`. The resulting domain is evaluated by `PercusYevickSolver`. Example workflows use `wavenumber="auto"`, which selects a zero-inclusive grid from the smallest particle radius and requested distance range. Users can instead supply a grid with `make_wavenumber_grid` when they need explicit control over radial resolution and Fourier sampling.

For explicit configurations, an RSA workflow combines a `PackingDomain`, a radius sampler, and `RSASOptions`. Calling `RSASimulator.run()` returns a `PackingResult`. The result contains centres, radii, class labels, packing statistics, and total or partial pair-correlation estimates. `MetropolisSimulator` accepts such a configuration with `MetropolisOptions` and reports displacement-move acceptance statistics. `PackingEstimator` reports ensemble correlation statistics together with insertion, rejection, runtime, and realised-packing diagnostics.

3. Illustrative examples

3.1. Complementary configuration workflows

Figure 2 shows a fixed-seed periodic RSA realisation of a binary mixture. The cross-section provides a direct geometric diagnostic of one generated configuration, while the correlation panel reports ensemble statistics for the same mixture. All quantitative figures are reproducible from the public API using the workflows distributed with the repository; the external reference data used for validation are versioned alongside the manuscript. Table 3

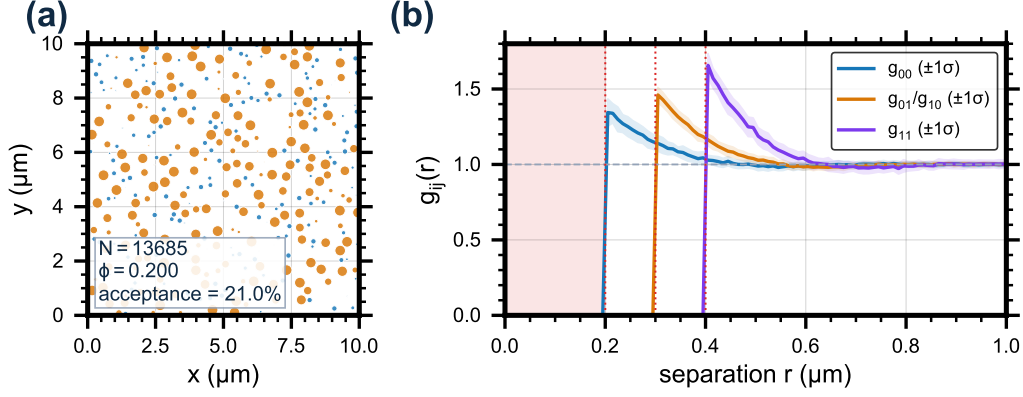


Figure 2: Reproducible periodic RSA demonstration for a two-radius mixture. The cubic domain has side length $10\text{ }\mu\text{m}$; sphere radii are 100 and 200 nm with requested number fractions 0.65 and 0.35; the target packing fraction is 0.20; and the random seed is 20260828. (a) Central cross-section of the accepted three-dimensional configuration, with colour indicating radius class. (b) The unique partial pair correlations, $g_{00}(r)$, $g_{01}(r) = g_{10}(r)$, and $g_{11}(r)$, estimated from 64 independent RSA realisations. The legend identifies each ensemble-mean coloured line; its matching same-colour band shows $\pm 1\sigma$, where σ is the sample standard deviation across realisations. The red shaded interval is the exclusion range shared by all pairs, while the red dotted lines mark the three contact separations, 0.20, 0.30, and 0.40 μm . The dashed horizontal line marks the uncorrelated limit $g_{ij}(r) = 1$.

117 collects the physical, stochastic, and numerical controls for all quantitative
 118 figures.

119 *3.1.1. Complementary RSA comparison and numerical convergence*

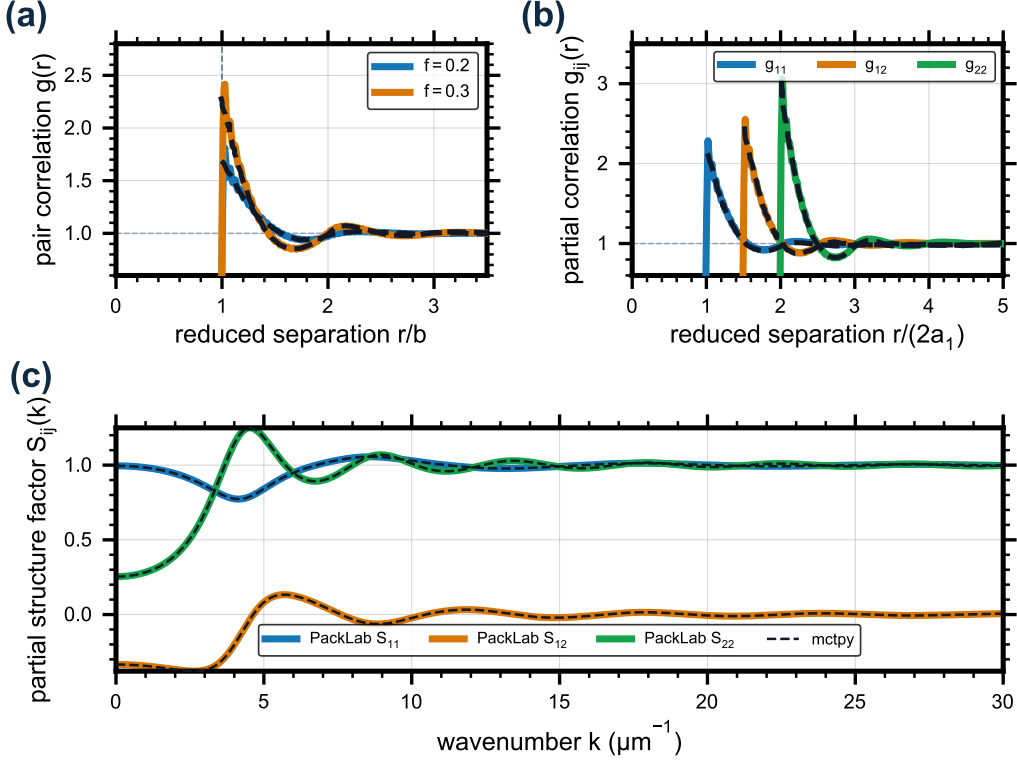


Figure 3: Independent validation of PackLab’s real- and reciprocal-space PY outputs. (a) Monodisperse PackLab correlations (coloured solid lines) and digitised Tsang et al. references [9] (black dashed lines) at volume fractions $f = 0.2$ and $f = 0.3$, plotted against r/b . (b) Binary-mixture PackLab correlations (coloured solid lines) and digitised references (black dashed lines) for $a_2 = 2a_1$ and partial volume fractions $f_1 = 0.2$, $f_2 = 0.1$, plotted against $r/(2a_1)$. (c) PackLab (coloured solid curves) and *mctpy* (black dashed curves) partial structure factors for radii 0.35 and 0.70 μm , number fractions 0.65 and 0.35, and total volume fraction 0.25.

120 3.2. Percus–Yevick validation and numerical resolution

121 3.2.1. Analytical correlation and structure-factor conventions

122 The RSA estimator is used only for the complementary comparison below.
 123 It defines $g_{ij}(r_b)$ as the ratio between the observed ordered i – j pair count in
 124 radial shell b and the ideal-gas expectation for the same configuration. For
 125 species counts N_i and N_j , domain volume V , and shell volume ΔV_b , the latter
 126 is

$$E_{ij,b} = N_i (N_j - \delta_{ij}) \frac{\Delta V_b}{V}, \quad g_{ij}(r_b) = \frac{C_{ij,b}}{E_{ij,b}}, \quad (3)$$

127 where δ_{ij} is the Kronecker delta and $C_{ij,b}$ is the observed ordered count. The
 128 radial bins are spherical shells with centres $r_b = (b + 1/2)\Delta r$ and extend
 129 to half the shortest box dimension; periodic estimates use minimum-image
 130 separations. When pair sampling is requested, the observed histogram is
 131 rescaled by the ratio of all unordered pairs to examined unordered pairs before
 132 this normalization.
 133 For the analytical workflow, $h_{ij}(r) = g_{ij}(r) - 1$ and $H_{ij}(k) = \sqrt{\rho_i \rho_j} \tilde{h}_{ij}(k)$ is
 134 the density-scaled reciprocal-space total correlation. We use the corresponding
 135 density-scaled partial structure-factor convention

$$S_{ij}(k) = \delta_{ij} + H_{ij}(k). \quad (4)$$

136 This convention makes both the pair-correlation and structure-factor com-
 137 parisons dimensionless and identifies the density factors required when trans-
 138 forming between them.
 139 The PY calculation uses an inverse radial Fourier transform, so its real-space
 140 result depends on wavenumber-grid extent and sampling density. For an
 141 isotropic mixture, the real-space total correlation is recovered from $H_{ij}(k)$
 142 through

$$h_{ij}(r) = \frac{1}{2\pi^2 \sqrt{\rho_i \rho_j}} \int_0^\infty k^2 H_{ij}(k) \text{sinc}(kr) dk. \quad (5)$$

143 At the maximum requested distance r_{max} , the $\text{sinc}(kr)$ factor has wavenumber
 144 period $2\pi/r_{\text{max}}$, which is the most rapidly varying kernel over the requested
 145 distance range. A truncated grid omits high-wavenumber content, whereas
 146 a coarse grid under-samples oscillations in the sinc kernel. Either case can
 147 produce a visually smooth but numerically misleading correlation function.
 148 For `wavenumber="auto"`, PackLab uses a real-space resolution of one twenti-
 149 eth of the smallest radius and twelve samples per sinc-kernel oscillation at
 150 the maximum requested distance. When a user-provided grid has fewer than
 151 eight samples per oscillation, PackLab raises a runtime warning. These are
 152 numerical defaults, not physical constants, and the grid-construction API
 153 allows stricter choices.

154 3.2.2. Independent analytical comparisons

155 Figure 3 compares PackLab with independent real- and reciprocal-space PY
 156 results. Panels (a) and (b) compare PackLab's mono- and binary-mixture
 157 solutions with manually digitised PY curves from Tsang et al. [9] at matched
 158 volume fractions and radius ratio. The expected hard-core contact locations,

159 decay toward the uncorrelated limit, and damped post-contact structure
 160 agree visually across both monodisperse curves and all three binary partial
 161 correlations. Because the reference coordinates were manually recovered from
 162 printed plots, their pointwise uncertainty is not sufficiently controlled for a
 163 meaningful numerical error metric. The comparison is therefore interpreted
 164 qualitatively and is used to check the principal physical features of the
 165 published solutions rather than to estimate implementation error.
 166 Panel (c) provides a higher-precision reciprocal-space check against versioned
 167 reference data generated with the independently implemented `hsmPY` compo-
 168 nent of `mctpy` version 0.0.2.post1 [8]. Both calculations use the density-scaled
 169 convention of Equation 4, and the checked-in reference contains 601 wavenum-
 170 bers from 0 to $30 \mu\text{m}^{-1}$. For the displayed binary mixture, the maximum
 171 absolute difference across all partial structure factors is 9.58×10^{-10} . The
 172 same reproducible comparison gives maximum differences of 2.32×10^{-11} and
 173 5.97×10^{-10} for the monodisperse and ternary cases, respectively.
 174 Figure 4 distinguishes model comparison from numerical verification. Panel (a)
 175 compares all unique partial correlations from a 100-realisation RSA ensemble
 176 with PY evaluated for the same binary-mixture inputs and the same radial
 177 coordinates. The differences are expected physical consequences of comparing
 178 finite, irreversible RSA with an equilibrium PY reference, rather than a
 179 numerical-error measure. Panel (b) compares PY with 12 independently
 180 initialized MMC chains after 600 burn-in sweeps. The mean move-acceptance
 181 rate is 0.782, and the post-contact root-mean-square (RMS) differences are
 182 0.045, 0.027, and 0.032 for g_{00} , g_{01} , and g_{11} , respectively. Between 300 and
 183 600 sweeps, 94.3% of resolved correlation values agree within the combined
 184 95% uncertainty of the two ensemble means. Because PY is approximate, this
 185 is an equilibrium-consistency check rather than an exact-error measurement.
 186 Panel (c) isolates numerical convergence by measuring the PY result against
 187 a separately specified, much finer wavenumber grid.

188 4. Impact

189 PackLab turns a collection of mixture definitions, analytical expressions,
 190 Fourier transforms, configuration generators, and statistical estimators into a
 191 reusable and inspectable research workflow. Its principal impact is method-
 192 ological reproducibility: a study can carry the same radii, composition, volume
 193 fraction, units, and reciprocal-space resolution from mixture construction
 194 to pair correlations, structure factors, and optional dependent-scattering

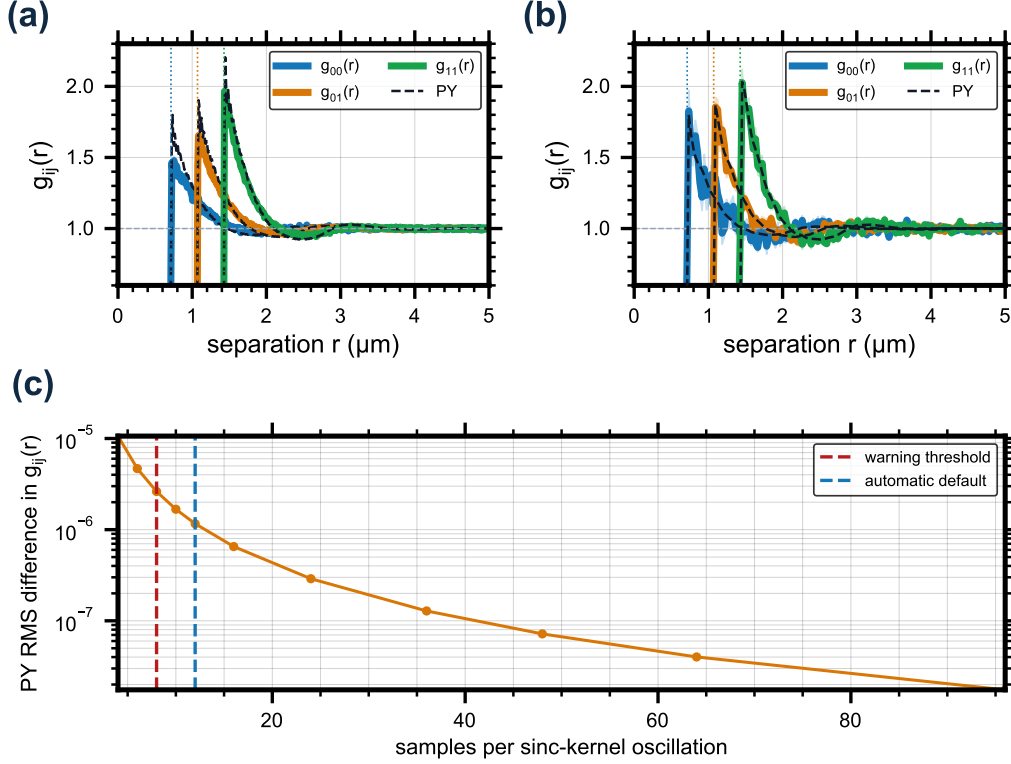


Figure 4: Matched configuration–PY comparisons and PY grid convergence for a binary mixture with radii 0.357 and 0.714 μm , equal number fractions, and total volume fraction 0.25. (a) RSA ensemble means (coloured solid curves) with one-standard-error bands and PY references (black dashed curves) for the three unique partial correlations; dotted lines mark hard-core contact distances. (b) Means from 12 independently initialized MMC chains after 600 burn-in sweeps, with one-standard-error bands and matched PY references. (c) RMS difference from a 512-samples-per-oscillation PY reference as the wavenumber grid is refined; red and blue dashed lines mark the warning threshold (8) and automatic default (12), respectively.

195 calculations. The Intralipid model of Attendu et al. [3] illustrates the class of
 196 research enabled by this workflow, in which measured particle-size informa-
 197 tion is combined with multicomponent hard-sphere correlations to evaluate
 198 concentration-dependent optical scattering. The public Python interface and
 199 executable examples also make systematic parameter sweeps and indepen-
 200 dent numerical checks accessible without requiring users to reimplement the
 201 Lebowitz mixture solution or its transforms.

Table 3: Reproducibility controls for the quantitative figures. Figure numbers refer to the rendered manuscript.

Figure	Mixture and geometry	Stochastic controls	Numerical and reference controls
2	Radii 100 and 200 nm; number fractions 0.65 and 0.35; periodic cube of side 10 μm ; target $\phi = 0.20$.	Seed 20260828; 64 independent realisations; 500 radial bins.	Pair correlations to 2 μm ; maximum attempts and consecutive rejections of 100000 and 20000.
3	Tsang mono- and binary-mixture inputs; <code>mctpy</code> mono-, bi-, and ternary-mixture checks.	Deterministic analytical calculations.	Automatic grids for $g(r)$; digitised curves assessed qualitatively; 601 points over $0 \leq k \leq 30 \mu\text{m}^{-1}$ for the quantitative $S_{ij}(k)$ check.
4	Radii 0.357 and 0.714 μm ; equal number fractions; $\phi = 0.25$; periodic cubes of side 30 μm (RSA) and 15 μm (MC).	RSA seed 2026, 100 realisations, and 1500 bins; 12 MC chains with seeds 4100–4111 and 5100–5111, 600 burn-in sweeps, and 250 bins.	MC displacement 0.12 μm ; grid resolutions 4–96 samples per oscillation; 512-samples-per-oscillation reference; $r_{\text{max}} = 10 \mu\text{m}$ and 1000 radial points.

202 4.1. Quality control and availability

203 PackLab tests analytical hard-core exclusion, automatic-grid selection, grid
204 warnings, and convergence through its public API. Separate tests cover sam-
205 plers, packing validity, periodic boundaries, and Metropolis move validation.
206 Executable documentation examples are organised around analytical, RSA,
207 MMC, and comparison workflows. Continuous integration (CI) builds the
208 native extensions and runs the test suite for CPython 3.11–3.13.
209 PackLab version 0.7.0 requires Python 3.10 or later, is released under the MIT
210 License, and is available from Python Package Index (PyPI) and conda. Source
211 code, documentation, issue tracking, and release history are available at <https://github.com/MartinPdeS/PackLab>.
212 Release-specific citation metadata is
213 maintained in `.zenodo.json`. Archived releases are available from the Zenodo
214 record at [10.5281/zenodo.20207810](https://zenodo.org/record/10.5281/zenodo.20207810).

215 4.2. Limitations

216 The current scope is three-dimensional, spherical, hard-core particles in
217 cuboidal domains. RSA should not be interpreted as an equilibrium-fluid

218 simulator; MMC requires explicit equilibration assessment; and PY should
 219 not be interpreted as a measured structure or a general interaction model.
 220 The practical limits of the stochastic workflows depend on the requested
 221 mixture and numerical settings. For RSA, insertion becomes progressively
 222 less efficient as the requested packing fraction increases; finite domains,
 223 broad radius distributions, and stringent stopping conditions can prevent the
 224 requested target from being reached. Consequently, users should report the
 225 realised packing fraction, stopping diagnostics, random seed, ensemble size,
 226 and domain size rather than treating a requested target as an achieved state.
 227 Pair-correlation estimates from finite configurations additionally depend on
 228 binning and are subject to finite-size uncertainty, particularly near the largest
 229 separations accessible in a domain. For MMC, users should assess burn-in,
 230 autocorrelation, proposal acceptance, and finite-size effects before treating
 231 separated chain states as equilibrium samples.
 232 For PY, the approximation error depends on density, composition, and particle-
 233 size ratio, and is separate from Fourier-grid discretisation error. The automatic
 234 grid and under-resolution warning are useful safeguards, but they do not
 235 establish convergence for every mixture: the default balances the maximum
 236 wavenumber, which controls real-space resolution, against the wavenumber
 237 spacing, which controls sampling of the inverse-transform kernel. Sharper
 238 near-contact structure or extreme size ratios may require a larger maximum
 239 wavenumber, whereas correlations over a longer distance range may require
 240 a smaller wavenumber spacing. Users should refine the wavenumber grid
 241 and compare against an independently chosen equilibrium benchmark when
 242 the analytical result informs a quantitative conclusion. PackLab’s separate
 243 namespaces, explicit input definitions, recorded stochastic settings, and grid
 244 diagnostics are designed to make these limitations visible at the point of use.

245 5. Conclusions

246 PackLab provides a unit-aware Python interface to multicomponent PY pair
 247 correlations and partial structure factors, supported by native C++20 kernels
 248 and explicit Fourier-grid diagnostics. Qualitative real-space comparisons
 249 recover the principal features of published hard-sphere-mixture curves, while
 250 reciprocal-space results agree with a separately implemented reference to
 251 within 9.6×10^{-10} for the tested mono-, bi-, and ternary mixtures. The
 252 package’s RSA generator and MMC sampler extend the workflow to explicit
 253 configurations without treating their outputs and the analytical reference as

interchangeable. Together, these facilities support reproducible structural inputs for hard-particle and dependent-scattering studies while retaining the numerical and physical assumptions needed to interpret each result.

CRedit authorship contribution statement

Martin Poinsinet de Sivry-Houle: Conceptualization, Methodology, Software, Validation, Visualization, Writing–original draft, Writing–review and editing. Xavier Attendu: Methodology, Validation, Writing–review and editing. Ton G. van Leeuwen: Supervision, Writing–review and editing. Edwin van der Pol: Supervision, Writing–review and editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data availability

All source code, tests, figure-generation scripts, and numerical reference data used in this article are available in the PackLab repository and its versioned release archive.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with manuscript organization, language editing, and software-maintenance tasks. The authors reviewed and edited the output and take full responsibility for the content of the publication.

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